Glock Safe-Action Pistols

Glock • Generations 1–5 (9x19mm, .40 S&W;, .45 ACP, 10mm Auto)

Manufacturer:	Glock	**Category:**	Pistols
Action Type:	Striker-Fired Safe Action	**Era / Service:**	Modern Duty & Tactical (1982–Present)
Calibers:	9x19mm, .40 S&W;, .45 ACP, 10mm Auto, .357 SIG	**Covered Models:**	Glock 17, Glock 19, Glock 21, Glock 22, Glock 26, Glock 34

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Cruciform Sear Check (66%) • 5-Drop Lube Protocol • 25¢ Trigger Polish • 30274 Ejector Bar Upgrade • Detail Strip

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
CRUCIFORM SEAR ENGAGEMENT	≥ 66% (minimum 2/3 sear overlap) verified via orange armorer inspection cover plate (SP 02096)
CONNECTOR GAP	0.005" to 0.010" clearance between connector top tab and trigger housing wall
TRIGGER PULL STANDARD	5.5 lbs (standard unmarked connector) / 4.5 lbs (dot) / 3.5 lbs (minus connector)
FIRING PIN PROTRUSION	0.045" to 0.055" through breech face; must retract freely under spring cups
SLIDE RAIL CLEARANCE	0.008" to 0.014" slide-to-frame rail runout
LUBRICATION POINTS	Exactly 5 drops: 1 each rear slide rail, 1 each front slide rail, 1 barrel exterior hood/lug, 0.5 connector ramp wipe
CRITICAL WARNING	DO NOT lubricate firing pin channel or firing pin safety plunger cavity; must remain 100% dry to avoid hydraulic lock and misfires

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Slide Out of Battery (1/16" to 1/8" Gap) **Slide hangs up just short of lockup; trigger cannot reset or fire**	• Carbon buildup on barrel hood or breech face locking shoulder. • Weak recoil spring assembly (captured guide rod fatigue). • Over-expanded reloaded case or fouled match chamber. • Deformed polymer guide rod rubbing slide dust cover channel.	**Immediate Action: Tap rear of slide with heel of hand to seat in battery. Gunsmith Action: Run Glock 45° recoil spring test (empty weapon, cock, pull trigger, point 45° straight up, ease slide rearward 1/4" and release gently; slide must go fully into battery under spring pressure alone). Replace RSA if failed.**
Stoppage 2: Erratic Ejection / Brass to Face (BTF) **Spent casing ejects straight back into armorer face or dribbles out of port**	• Early Gen 4 336 stamped ejector bar installed instead of updated 30274/47021 part. • Chipped or deformed extractor claw face. • Extractor depressor plunger loaded backwards (metal bearing must touch extractor, polymer bearing touches slide cover plate). • Excessive carbon or brass shavings packed in extractor claw pocket.	**Gunsmith Action: Inspect extractor claw under 10x loupe. Upgrade ejector bar to factory Gen 4 30274 (9mm) or Gen 5 47021. Verify extractor plunger spring order: plunger -> spring -> polymer spring-loaded bearing.**

Stoppage 3: Light Primer Strikes (Click Instead of Bang) **Firing pin releases but indentation on primer cup is shallow or off-center**	• Oil, solvent, or hardened carbon inside firing pin channel liner causing hydraulic drag. • Chipped or worn firing pin tip. • Cracked or broken polymer firing pin spring cups (SP 00070). • Aftermarket reduced-power striker spring installed (4.0 lb vs factory 5.5 lb).	**Gunsmith Action: Detail strip slide. Inspect firing pin channel; clean with dry cotton swab (NEVER oil). Shake slide back and forth; striker must click and slide freely. Replace spring cups if cracked.**
Stoppage 4: Trigger Fails to Reset (Dead Trigger After Cycling) **Slide cycles rearward, extracts and chambers next round, but trigger remains slack against rear frame**	• Broken or unhooked coil trigger spring in trigger housing (NY1/NY2 spring inverted). • Connector bent inward too far (connector lip dragging heavily inside housing cut). • Trigger bar cruciform binding on frame ejector housing. • Cruciform drop safety shelf dropped below ejector pin ledge.	**Gunsmith Action: Check connector angle. Bend connector slightly outward until a sheet of paper (approx 0.008") passes between connector top lip and trigger housing. Verify trigger spring 'S' hook orientation.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Glock Complete Detail Disassembly & Reassembly Masterclass	MidwayUSA / Larry Potterfield	Step-by-step frame and slide complete teardown using standard 3/32" armorer punch. Removing locking block, trigger mechanism housing, connector, and striker channel liner.
Glock 25¢ Trigger Job & Cruciform Sear Safety Engagement Check	Brownells Gunsmithing Bench	Polishing bearing surfaces (trigger bar tab, connector ramp, safety plunger, cruciform sear shelf) with Mother's polish. Verification of 66% sear engagement with armorer plate.